4.5.2.7 X-47B (2012)

The Northrop Grumman X-47B, first unveiled in 2012, represents one of the most groundbreaking milestones in autonomous combat aviation and the evolution of third-generation to fourth-generation unmanned combat aerial vehicles (UCAVs). Developed under the US Navy's unmanned combat air system demonstration (UCAS-D) program, the X-47B was designed to explore the integration of unmanned, carrier-based operations, a domain previously limited to manned aircraft. It was the first UAV in history to successfully perform autonomous take-off and landing on an aircraft carrier, marking a major leap toward the realization of fully autonomous air combat operations in naval environments (Tekinalp, et al. 2012).

Design and technical overview

The X-47B employs a tailless, blended-wing-body configuration, optimized for stealth, endurance, and aerodynamic efficiency. Constructed using composite materials and radar-absorbent coatings, its design significantly reduces radar cross-section (RCS), ensuring survivability in contested airspace. The aircraft measures approximately 11.6 meters (38 feet) in length, with a wingspan of 18.9 meters (62 feet), which can be folded for carrier storage and deck maneuvering.

Powered by a Pratt and Whitney F100–220U turbofan engine, the X-47B achieves a cruise speed of 0.9 Mach and operates at altitudes up to 40,000 feet (12,200 meters). Its internal weapons bays can accommodate up to 2000 pounds (900 kg) of precision-guided munitions, such as the GBU-31 JDAM or GBU-12 Paveway II, maintaining its stealth profile during strike missions.

The airframe's smooth contours and the absence of vertical stabilizers reduce radar reflectivity, while its autonomous flight control system enables